MILITARY SPECIFICATION

CRANE, HYDRAULIC, TRUCK MOUNTED, A/S32A-16 16,500 POUND CRANE CAPACITY, 6X4, DED

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a diesel-engine-driven, 6X4, truck mounted, fully revolving, hydraulically operated crane of 16,500 pounds capacity, designated A/S32A-16.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-P-514	Plate, Identification, Instruction and Marking, Blank
MIL-C-3580	Cranes and Crane-Shovels, Truck, Crawler, and Wheel Mounted, Full Revolving, and their Attachments, Packaging of
MIL-H-5606	Hydraulic Fluid, Petroleum Base, Aircraft and Ordnance
MIL-M-8090	Mobility, Towed Aerospace Ground Equipment, General Requirements for
MIL-A-8421	Air Transportability Requirements, General Specification for

Federal

KKK-T-2109	Truck and Truck-Tractors, Commercial, Diesel and Gasoline Engine Driven, 34,500 to 64,000 Pounds GVW, 6X4

STANDARDS

Federal

FED-STD-595	Colors

FSC 3810

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Military

DOD-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the selection of
MIL-STD-808	Finishes, Protective, and Codes, for Finishing Schemes for Ground and Ground Support Equipment
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Laws and Regulations:

40 CFR 86	-Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines.
40 CFR 205	-Noise Emission Standards for Transportation Equipment and Heavy Trucks.
49CFR 393	-Federal Motor Carrier Safety Regulations (FMSCR).
49CFR 571	-Federal Motor Vehicle Safety Standards (FMVSS).

(The code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

Society of Automotive Engineers, Inc. (SAE)

J534	Lubrication Fittings
J588	Turn Signal Lamps
J589	Turn Signal Switch
J682	Rear Wheel Splash and Stone Throw Protection
J931	Hydraulic Power Circuit Filtration

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(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale PA 15096.)

State of California

Vehicle Code of California

(Application for copies should be addressed to the Department of Motor Vehicles, 2570 24th Street, Sacramento CA 95809.)

The Tire and Rim Association, Inc.

Yearbook

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 W Market Street, Akron OH 44313.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using federal agencies.

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing (see sections 4 and 6).

3.2 Components. The crane shall consist of at least the following major components:

Item or System Name	See Requirement
Chassis	3.7.1
Outriggers	3.7.2
Suspension lockouts	3.7.3
Crane base frame	3.7.4
Crane power source	3.7.5
Swing machinery	3.7.6
Load hoisting machinery	3.7.7
Boom hoisting machinery	3.7.8
Boom	3.7.9
Operator's cab	3.7.10
Operator's control	3.7.11
Electrical system	3.7.12

3.2.1 Description. The crane shall be a commercial model consisting essentially of a six-wheel, 4-rear-wheel drive, front-wheel steer, pneumatic-tired, diesel engine powered carrier. The carrier shall be equipped with a full revolving superstructure with cab, a basic hydraulic telescoping boom of no more than 30 feet radius and capable of extending to a radius of not less than 40 feet when the boom is in a horizontal position. The maximum overall width of the crane shall not exceed 96 inches, excluding the tire bulge. The overall height, with crane boom in travel position, shall be not more than 107 inches with a maximum overall length of not more than 30 feet. The gross weight shall not exceed 42,000 pounds.

3.3 Selection of standards and specifications. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

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3.4 Materials. Recycled and recovered raw materials should be used to the maximum extent possible in lieu of virgin raw materials as long as these materials do not jeopardize the intended use and fully comply with all contract requirements. Materials used shall be free from defects which would affect the performance or maintainability of individual components of the overall assembly.

3.4.1 Thermal insulation material. All insulating material shall maintain its shape and not settle, sag, pack, abxorb water, or disintegrate under the vibrations and temperatures encountered in use. All insulation shall be fire resistant.

3.5 Design and construction. Where no specific construction requirements are specified herein, the manufacturer's best commercial practice shall apply. All parts, components, and attachments shall be identical for each crane furnished under any specific contract. No deviations will be accepted without prior written approval of the contracting officer. The shift pattern of the truck chassis shall not be changed.

3.5.1 Stress requirements. Design of the completely assembled crane shall be such that the sustained maximum unit stress imposed by the rated load shall be not more than 60 percent of the yield point for any structural member.

3.5.2 Reliability. The crane shall have a minimum mean-time-between-failure (MTBF) of 43 hours with a minimum reliability of 0.98 for a one-hour mission at a confidence factor of 0.90 (see 6.3.1).

3.5.3 Maintainability. The crane design, together with component and accessory location and installation, shall permit ready accessibility to all items requiring period maintenance service in the field and provide the following:

a. Minimum number of parts consistent with reliability and performance specified herein.

b. Minimum amount of training and time necessary for assembly, disassembly, location of trouble sources, and maintenance including servicing. Where practical, parts and components shall be located or positioned for rapid and simple inspection and recognition of excessive wear or potential failure.

c. Permit adjustments, servicing, replacement of parts and components, and other maintenance with minimum distrubance to other equipment part or components. Parts and components shall be located for ample and rapid access unless performance or reliability will be appreciably degraded by the accessible location. In performing maintenance, if engineering reasoning or data determines that physical or visual interference between items cannot be avoided, the item predicted to require the most maintenance shall be located for best accessibility.

d. Permit maintenance with general-purpose tools and equipment normally available commercially. Use of special-purpose tools and equipment shall be subject to approval by the procuring activity.

e. Minimum number of tools required for maintenance by design practices such as reducing the variety of bolt head sizes to the practicable minimum.

f. Where possible, intricate locking devices, controls, and threading fastenings that can be easily over-torqued by operators lacking feel through thick gloves or numbness shall be avoided. Covers or plates that must be removed for component adjustments or for parts removal shall be equipped with substantial, quick-disconnect fastenings.

3.5.4 Foolproofness. Where improper installation of an item could cause malfunction of the item or the equipment in which it is installed, an unsymmetrical mounting means shall be provided. The mounting shall allow the item to be installed only in its proper operating position. If an unsymmetrical mounting means is not practical, the item shall be so mounted that its proper operating position can be readily and visibly determined by service maintenance personnel.

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3.5.5. Safety features.

3.5.5.1 Guards. All rotating or reciprocating parts and all parts subject to high operational temperatures or electrically energized that are of such nature or are so located as to be or become a hazard to the operating or attending personnel, shall be substantially guarded or insulated to the extent necessary to eliminate the hazard. The principal platform walking surfaces shall be of an antiskid type. Ladders, steps, and handholds shall be provided in such quantity and in such size on the sides of the carrier, carrier cab, and superstructure that entrance thereto and exit there from may be unhampered and nonhazardous. Engine cooling fans shall have heavy grille or ring type guards.

3.5.5.2 Boom capacity indicators.

3.5.5.2.1 Boom angle indicator. A pendulum type boom indicator shall be provided and so mounted as to be easily read in the crane cab..

3.5.5.2.2 Boom length indicator. A boom length indicator shall be provided for the crane operator to readily determine the boom length.

3.5.5.2.3 Boom capacity chart. A boom capacity chart shall be provided in the crane cab to be used in conjunction with the boom length indicator and boom angle indicator and boom angle indicator. This chart shall be located near and adjacent to boom angle and boom length indicators.

3.5.5.3 Safe load control device. A device shall be provided to limit the crane's lifting operations to loads of not greater than 85 percent of tipping loads at all radii, both with and without use of the outriggers. This device shall govern the crane operations to prohibit overloading at all radii, with minimum to maximum boom lengths, by stopping the boom hoist in a lowering direction and the load hoist in a hoisting direction when the maximum safe load-radius combination is reached. Visual and audible warnings shall be provided to warn the operator of the approach to the safe load limits. The safe load control device shall cut off the engine when the fuse is blown or removed.

3.5.5.4 Deadman control and automatic brakes. With manual release of any craning motion, the operator's control shall automatically apply brakes to the motion, immediately stopping and restraining the motion within rated load conditions. Each boom actuating mechanism shall be provided with safety features to prevent descent of the load in the event of power failure or mechanism failure of the hoist line mechanism, boom angle mechanism, or boom length mechanism. An additional locking device shall be provided to automatically prevent change of boom angle in the event of boom angle mechanism failure.

3.5.5.5 Hoist system failure. Means shall be provided to prevent injury to the operator in the event of breakage of either the boom hoist system or the load hoist cable.

3.5.6 Lubrication. The crane mechanism shall be designed and constructed to operate satisfactorily when lubricated with the standard military lubricants specified in MIL-M-8090.

3.5.6.1 Lubrication fittings. Lubrication fittings shall conform to SAE J534 and shall be located in accessible, protected positions. Extended fittings shall be provided to lubricate parts or assemblies that are not readily accessible for direct lubrication or which are likely to be overlooked because of inaccessibility.

3.5.6.2 Lubrication chart. A lubrication chart shall be provided directing attention to all lubrication fittings and specifying the range and grade of lubricant required for critical temperatures. The chart shall be permanently attached to the unit in an accessible and convenient location.

3.5.7 Steps or ladders. Steps or ladders shall be provided and properly located to provide safe and easy access to both the truck and crane.

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3.5.8 Air transportability. The crane shall be designed for air transportability in the assembled condition in a Type C-141 aircraft in accordance with MIL-A-8421.

3.6 Performance.

3.6.1 Environmental conditions. The crane shall operate without winterization kits in ambient temperatures ranging from +125° to -20°F..

3.6.2 Mobility. When fully equipped, the crane shall attain the performance on dry, level concrete or asphalt surfaces, as specified in KKK-T-2109.

3.6.3 Lifting capacity. With the outriggers extended, the crane shall be capable of handling a 10,000 pound load with the load at a radius (see 6.3.7) of 20 feet, lifting and lowering the load vertically at speeds up to 45 fpm. Without the outriggers extended, the crane shall be capable of lifting a 5,000 pound load at a radius of 20 feet, lifting the load vertically until the hook is 40 feet above the ground. With these loads suspended just above the ground, the crane shall be capable of rotating 270° at a speed up to but not to exceed 4 fpm. The revolving motion shall be capable of stopping at any position relative to the chassis. The distance between the hook and the chassis or outriggers shall be at least 9 feet 6 inches during this rotation. For precision operation, the crane shall permit hoisting and lowering of all rated loads at a speed not to exceed 2 fpm. Free spooling of line will not be permitted. Operation at any given hoisting or lowering speed and at varying speeds shall provide smooth and steady movement of a load. Under any of these conditions, no component of the crane shall exceed 60 percent of its design yield strength, and the crane shall not exceed 85 percent of its tipping load.

3.6.3.1 Transporting capacity. The crane shall be capable of transporting a 10,000 pound load, 20 feet in length, 8 feet in height, and 8 feet in width for 100 feet on a level concrete road.

3.7 Details of components.

3.7.1 Chassis. The chassis shall be a 6 by 4 vehicle conforming to KKK-T-2109, type I, class as required. The crane base shall be mounted thereon for support of the entire crane and loads.

3.7.1.1 Additional truck requirements. The following items as offered and described in KKK-T-2109 shall be provided:

a. Heavy duty cooling system.

b. Frame reinforcement for type I vehicle.

c. A single cab mounted along the left side of the truck engine.

(The crane boom shall be positioned over the truck engine for transporting.)

3.7.1.2 Exceptions to truck requirements. The following items as offered and described in KKK-T-2109 shall not be provided:

a. Spare wheel or rim.

b. Spare wheel carrier.

c. Spare tire.

d. Tools.

e. Radio interference supression in accordance with paragraph 3.4.2.9 (the entire vehicle shall be radio interference supressed in accordance with paragraph 3.9 of MIL-C-27718).

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3.7.1.3 Splash shields and mud flaps. The rear wheels shall be equipped with splash shields in front and mud flaps at the rear.

3.7.2 Outriggers. Four power operated outriggers shall be provided, one on each side at the rear and one on each side forward of the crane base, located to provide stability when lifting maximum loads. Outriggers shall be adaptable to manual raising, lowering, and locking, and shall not extend beyond the width limits of the chassis when in the raised position. The warning plate for the outrigger switch shall be located near the switch.

3.7.3 Suspension lockout devices. Readily actuated or adjustable suspension lockout devices shall be furnished for each spring to provide maximum stability during operation of the crane. The suspension lockouts shall be removable and replaceable so that spring and shock absorber maintenance and repair is not degraded.

3.7.4 Crane base frame. The crane base frame shall be of heavy structural steel members rigidly fastened to the chassis frame.

3.7.4.1 Bearing swing gear circle. A bearing swing gear circle shall be fastened to the crane base. It shall support the revolving crane frame by means of large diameter, double row ball bearings or equal. The bearings and raceway shall be packed in grease and protected by a grease labyrinth seal. The swing gear circle shall transmit all crane loads to the crane base.

3.7.4.2 Revolving frame. The revolving frame shall be of heavy structural steel members rigidly fastened to the upper portion of the bearing swing gear circle. It shall be properly shaped and constructed to mount and carry the loads of all components to be mounted thereon.

3.7.5 Crane power source. Power for all crane motions shall be provided by either an electrical generator or a hydraulic pump driven by a separate engine. If electric, a generator with regulator, motors, and other items for a complete power system shall be provided. If hydraulic, a pump, pressure regulator, and reservoir of sufficient capacity for complete displacement on all systems, controls, oil cooler, and other items for a complete power system shall be provided. The hydraulic system shall use hydraulic oil conforming to MIL-H-5606. A hydraulic fluid filter shall be installed in the pump suction line to assure a clean supply of oil to the hydraulic pump. Filtration shall be adequate to prevent materials from clogging valves, ports, or controls in the hydraulic system. An oil level indicator shall be provided for hydraulic oil reservoir.

3.7.5.1 Crane engine. The crane engine shall be a current standard, internal combustion, diesel type. The engine shall drive the crane power source. The engine shall drive the crane power source. The engine shall produce the required crane performance when operated on fuel of regular grade at the engine speed not in excess of that recommended by the engine manufacturer.

3.7.5.2 Governor. An overspeed governor shall be provided and set to limit engine speed to the maximum operating speed recommended by the engine manufacturer.

3.7.5.3 Engine lubrication system. The engine lubricating system shall provide adequate lubrication under all operating conditions and shall be equipped with an oil pump and an oil filter conforming to the size recommended by the engine manufacturer.

3.7.5.4 Cooling system. The engine cooling system shall be the manufacturer's current standard, heavy-duty type and of adequate capacity and design to efficiently control the temperature of the engine when operating at an ambient temperature of +125°F to -20°F (see 3.6.1).

3.7.5.5 Air filter. The engine shall be provided with the manufacturers standard heavy-duty, paper or oil-bath type air filter with a minimum oil capacity of 1 quart.

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3.7.5.6 Fuel system. The fuel system shall be the manufacturer's standard and separate from the truck fuel system.

3.7.5.6.1 Fuel tank. The fuel tank shall be of sufficient capacity for at least 8 hours continuous crane operation and shall be provided with gage.

3.7.5.6.2 Fuel filter. A fuel filter with filtering element of suitable material shall be mounted in a position where self-cleaning action will be effective where freezing of water which may collect will not impair its functioning and will be accessible for removal.

3.7.5.6.3 Fuel lines. The fuel lines shall be of suitable size tubing, solidwall type, equipped with standard fittings. The fuel lines shall have flexible hose at each point where vibration or flexure might cause tubing failure.

3.7.5.7 Exhaust system. The crane engine shall be equipped with an efficient muffler and exhaust system. The muffler and tailpipe shall be so located that exhaust gases will not enter the cab when the crane is operating under conditions specified herein. The exhaust piping shall be so installed that the cab temperature is not affected, and the outlet is at least 18 inches above the ground.

3.7.5.8 Air pollution control. The crane engine shall comply with 40CFR 86 governing control of air pollution from new motor vehicles and new motor vehicle engines applicable on the date of manufacture to non-military, non-export motor vehicle products. Vehicles with California destinations shall additionally comply with State of California regulations governing air pollution control applicable on date of manufacture.

3.7.6 Swing machinery. The crane swing motion provided through a ring gear and pinion shall be powered by either a variable voltage electric or hydraulic motor. The crane shall be capable of 360° continuous rotation.

3.7.7 Load hoist machinery. The load hoist machinery shall be mounted on the rotating frame. The capacity of the load hoist drum shall be such that all cable required for 4-part line operation with the maximum boom length can be spooled in two layers when reeved for a 2-part line operation with the minimum boom length. Sufficient cable for 4-part line operation with the maximum boom of the load hoist line shall be assured even when no load is imposed on the hook block. Adjustable limit switches shall be employed to prohibit overwinding the load hoist cable which would cause damage to boom point sheaves and hook block.

3.7.8 Boom hoist machinery. The boom hoist machinery shall be mounted on the rotating frame. Utilizing an electric motor hydraulic means, the boom hoist machinery shall move the hook to a shorter than maximum working radius with any load, not exceeding the rated capacity imposed on the hook. If a reeved system is provided, the capacity of the boom hoist drum shall be such that all cable can be spooled in two layers when reeved with the minimum boom length. The drum shall be grooved, and tight, level winding of the boom hoist line shall be assured. Adjustable limit controls shall be employed to prohibit overwinding the boom hoist cable which would cause damage.

3.7.9 Boom. A self-supported cantilever mounted boom, with a power-operated telescopic extension having a working capacity of at least 10,000 pounds, shall be provided. The boom shall be extendable from 30 to at least 40-foot lengths and shall be capable of handling the loads under the conditions specified in 3.6.3. A load block and safety hook assembly suitable for 1-, 2-, 3-, or 4- part hoise line operation shall be provided and so arranged that the assembly can be readily installed on or removed from the hoist line. The boom point shall have adequate sheaves and anchors to permit 1-, 2-, 3-, or 4-part hoist line operation. The boom shall be so located or stopped to prohibit it from striking the operator's cab in any operation of the boom hoist motion. Means shall be provided to prevent side sway of the boom tip during crane travel. It shall be possible to remove the boom by removing it from the revolving frame or by removing the revolving frame, with boom attached, from the chassis. A 4500 psi holding valve shall be provided for each hydraulic boom hoisting cylinder.

3.7.9.1 Boom limit controls. Automatic self-resetting limit controls shall be provided to limit the maximum and minimum boom angles.

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3.7.10 Operator's cab. The operator's cab shall be a one-man type of steel construction. The cab shall be fully enclosed and adequately insulated to provide inside temperatures not to exceed the outside temperature by more than 15° at a range of 75° to 125°F. The windshield and windows shall enclose all four sides of the cab and shall be so installed as to provide at least a 270° arc of vision about the crane at the operator's eye level. The cab shall contain the controls specified in 3.7.11. The operator's seat shall face the load with the boom in any position. All windows and doors shall be adequately weather stripped. The cab window on at least two sides shall be readily opened for ventilation. The operator's comfort and accessibility to the various controls and gages shall be considered in the design and layout of the cab. The cab shall be equipped with a sun-visor, cushioned, adjustable seat, dome light, personal heater and defroster, and electrically-powered windshield wipers on the front and rear windows. All windows shall be provided with safety glass.

3.7.10.1 Cab heater. The cab heater shall be the manufacturer's standard type. The heater shall provide and maintain a minimum cab temperature of 45° F at an outside temperature of -20° F. The heater shall be provided with the necessary controls and air distribution provisions to efficiently defrost the windshield.

3.7.11 Operator's controls. All controls for outriggers, boom hoist, load hoist, and rotation shall be hand-controlled and effective within safe limits. The controls shall be readily accessible and forward of the operator when seated in the normal operating position. The operator's control console shall also include at least the following: electric engine starter switch, ammeter, engine hourmeter, engine temperature and oil pressure gages, electric fuel gage, air pressure gage (if air is used) and switches for boom tip marker lights, radius indicator lights and flood lights. The crane controls and related instruments shall be grouped separately from the engine controls and related instruments.

3.7.12 Electrical system. All electrically energized operator's controls and lighting systems shall be operated on a 12-V electrical system. All electrical equipment shall be tropical insulated.

3.7.12.1 Lights. The crane shall be equipped with lights as follows:

a. Floodlights that will always face the direction of the boom and illuminate the working area. The floodlights shall provide a minimum intensity of 3 foot-candles over all load heights at maximum boom lengths.

b. Boom tip marker lights.

c. Dome light located in the crane operator's cab.

d. Boom capacity indicator light(s) as required.

3.8 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto, except for chassis.

3.9 Electromagnetic interference. The crane shall comply with the requirements specified in MIL-STD-461 for Class C3 equipment.

3.10 Weight. The crane shall be of minimum weight consistent with the requirements specified herein. The gross weight of the unit shall not exceed the gross vehicle weight rating of the chassis. The gross weight shall include the weight of the chassis, revolving crane, all accessories with fluid tanks full, and a load of 5,000 pounds.

3.10.1 Axle loading. Tandem axles spaced between 42 and 90 inches from the nearest adjacent axle shall not exceed an axle loading of 16,000 pounds each.

3.11 Finishes and protective coatings. Cleaning, painting, plating and anodic and chemical films shall be in accordance with MIL-STD-808. The color designation shall be DG. The engines shall be painted in the manufacturer's commercial standard color.

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3.12 Markings

3.12.1 Transportation data plates. Transportation data plates conforming to MIL-P-514 shall be provided.

3.12.2 Radio interference instruction plate. The following information shall be provided on metal instruction plates conforming to MIL-P-514. The instruction plates shall be installed in a conspicuous location on the dash panel of the cab.

CAUTION

THIS VEHICLE IS EQUIPPED WITH RADIO INTERFERENCE SUPPRESSION DEVICES. CARE SHOULD BE EXERCISED DURING MAINTENANCE TO RETAIN AND REINSTALL THESE DEVICES PRIOR TO REUSE OF UNIT.

3.13 Identification of product. Equipment, assemblies and parts, except the chassis, shall be marked for identification in accordance with MIL-STD-130.

3.14 Workmanship. The crane, including all parts and accessories, shall be constructed, assembled, and finished in a workmanlike manner. Particular attention shall be given to accuracy of dimensions; fit of panels and access doors; alignmnet of parts and assemblies; thoroughness of welding, brazing, soldering and riveting, and freedom from blemishes and defects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the crane shall be classified as follows:

a. Preproduction testing (see 4.3).

b. Acceptance testing (see 4.4).

4.3 Preproduction testing (see 6.2)

4.3.1 Test sample. One crane shall be subjected to the test specified in 4.3.3.

4.3.2 Test report. After completion of the preproduction tests, a test report shall be prepared in accordance with MIL-STD-831 and three copies of the report furnished the procuring activity.

4.3.2.1 Reliability and manintainability information. The following information shall be included as an appendix to the test report:

a. All failures, servicing, adjustments, maintenance, and irregular functioning shall be identified by accumulated operating time, cycles, miles, or position in the test procedure, as appropriate. Test conditions at the time of the events identified shall be recorded.

b. A summary of the engineering analysis and of any tests conducted to determine assignable causes for any failure or irregular functioning.

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c. A summary of the engineering analysis leading to any corrections made to design, construction, quality control, or other procedures, or leading to any corrections to be made or proposed to be made to production items. The summary shall also include an analysis of the predicted effectiveness of such corrections.

d. Clock time and man-hours required for each maintenance and servicing action taken during the tests. A brief description of the experience and qualifications of the personnel taking such actions shall be included.

e. Test activity or contractor comments on item features or requirements that, if modified, should improve the item.

f. Test activity or contractor comments on use or maintenance conditions to be avoided or cultivated to increase the reliability or useful life of the item.

g. Any of the above information that is already included in the test report body need not be repeated in the information required by this paragraph, but clear reference to the location of the data shall be included.

4.3.3 Preproduction tests. The preproduction test shall consist of all tests described under 4.5.

4.4 Acceptance tests. The acceptance tests shall consist of the following:

a. Individual tests.

b. Sampling plan and tests.

4.4.1 Individual tests. Each crane shall be subjected to the following tests as described under 4.5.

a. Examination of product.

b. Road test.

4.4.2 Sampling plan and tests

4.4.2.1 Periodic sampling. One crane shall be selected at random from each 50 or fraction thereof produced and subjected to the tests described in 4.5.3.1 through 4.5.3.3.

4.4.2.2 Rejection and retest. When an item selected from the production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all tests shall be repeated.

4.4.2.2.1 Individual tests may continue. For production reasons, individual tests or other sampling plans may be continued pending the investigation of a sampling test failure. But final acceptance of the items on hand or produced later shall not be made until it is determined that all items meet all the requirements of the specification.

4.4.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all the defects likely to be found and the method of correcting them.

4.5 Test methods.

4.5.1 Examination of product. The crane shall be inspected to determine compliance with the requirements of this specification with respect to materials, dimensions, markings and workmanship.

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4.5.2 Road test. The crane shall be driven at speeds varying from minimum to maximum for a total of 10 miles. During this time, the brakes, vehicle hydraulic system (when provided), and electrical components shall be operated.

4.5.3 Crane operations

4.5.3.1 Design overload. The crane utilizing the outriggers set shall be mechanically and firmly anchored to restrain it from tipping. In this condition, it shall lift 16,500 pounds to a vertical height of 2 feet from the ground using a boom radius of 20 feet with the boom at maximum length as determined by the safe load control, and revolve 270° then lower to the ground. After the load is removed, there shall be no signs of permenent deformation of crane components.

4.5.3.2 Tipping overload. The crane, in mobile order without using outriggers, shall lift 5,750 pounds to a vertical height of 2 feet from the ground using 20 foot radius with the boom at maximum length, as determined by safe load control, and revolve 270° without tipping.

4.5.3.3 Endurance. The crane shall operate through the following cycles 500 times each. Only minor mechanical adjustments shall be permitted during the tests. At the conclusion of the test, inspection shall reveal no undue wear or damage, but should indicate that the tests could have continued:

a. With the outriggers extended, a 10,000 pound load, at a radius of 20 feet, with the boom at maximum length as determined by the safe load control shall be lifted vertically until the hook is 20 feet above the ground. With this load lowered just above the ground the crane shall be rotated 270° and the load lowered to the ground.

b. With the outriggers retracted, a 5000 pound load, at a radius of 20 feet with boom at maximum length as determined by the safe load control shall be lifted vertically until the hook is 40 feet above the ground. With this load lowered just above the ground, the crane shall be rotated 270° and the load lowered to the ground.

c. A 10,000 pound load, 20 feet in length, 8 feet in height, and 8 feet in width, shall be lifted from the ground and transported for 100 feet on a level concrete road. The load shall be lowered to the ground.

4.5.3.4 Safety features. The crane shall be tested to show compliance with the safety features specified in 3.5.5.5. With the outriggers extended, a 10,000 pound load at a radius of 20 feet with the boom raised shall be lifted 2 feet above the ground. The load shall not descend more than 1 inch in a 24 hour period under the following conditions:

a. Operated continuously for a minimum of 4 hours under full operating load conditions immediately preceding test.

b. Vehicle engine stopped.

c. Boom hoist machinery, electric power line, or hydraulic line disconnected.

d. Load hoist machinery drive mechanism disconnected.

e. Boom extension mechanism or power source disconnected (electric line if electrically operated or hydraulic line if hydraulically actuated).

4.5.4 Environmental tests. The crane shall be subjected to the following environmental tests conducted in accordance with the specified procedures of MIL-STD-810.

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4.5.4.1 High temperature operation. The crane shall be subjected to high temperature in accordance with method 501, procedure I, except the exposure period shall be 24 hours at a maximum temperature of 125°F. At the end of the exposure period, the crane engine shall be operated at rated engine speed and operating horsepower until the coolant temperature of the engine has stablized at a constant temperature to insure compliance with the requirements specified herein. Crane motions as specified in 3.6.3 shall be accomplished. The inside cab temperature during this test shall not exceed 15°F of the outside or chamber ambient temperature.

4.5.4.2 Low temperature operation. The crane shall be subjected to low temperature in accordance with method 502, procedure I, except the exposure period shall be 24 hours at a temperature of -20°F. At the end of the 24 hour stabilization period, the crane engine shall be operated at rated engine speed and operating horsepower until the coolant temperature of the engine has stabilized at a constant temperature to insure compliance with the requirements specified herein. Crane motions as specified in 3.6.3 shall be accomplished. The inside cab temperature of 45°F or more shall be provided and maintained during this test. The window defrosting system shall maintain clear vision over at least 75 percent of the window area with the operator in the cab for 1 hour.

4.5.5 Electromagnetic interference suppression test. The crane and truck chassis shall be tested in compliance with the requirements of paragraph 3.9. This test shall be conducted with all electrical, mechanical and hydraulic systems operating.

4.5.6 Reliability demonstration and test. Satisfactory completion of all tests specified herein without failure demonstrates compliance with the quantitative requirements of this specification.

4.5.7 Maintainability evaluation. The crane shall be inspected and evaluated from the standpoint of ease of maintenance, servicing, and operation. Particular attention shall be given maintenance with a minimum number of common tools, servicing, and operation by personnel wearing heavy glooves, and provisions made to prevent accumulation of dirt, snow, ice, et cetera, that may hinder servicing and operation.

4.6 Inspection of preparation of delivery. Preservation, packaging, packing, and marking shall be inspected to verify conformance to the requirements of section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. The crane shall be preserved and packaged in accordance with MIL-C-3580.

5.1.2 Level C. The basic unit, component parts, repair parts, attachments, and component parts groups, tools, and publications shall be preserved and packaged to afford protection against deterioration and damage.

5.2 Packing. Packing shall be level A or C, as specified (see 6.2).

5.2.1 Level A. The crane shall be packed in accordance with MIL-C-3580.

5.2.2 Level C. The basic unit shall be shipped unboxed. Component parts, attachments, repair parts, component parts groups, tools, and publications shall be packed to insure carrier acceptance and save delivery to destination at lowest rates.

5.3 Marking. Packages, bundles, basic unit, attachments, and shipping containers shall be marked in accordance with MIL-STD-129. Teh nomenclature shall be: Crane, Truck Mounted, A/S32A-16.

6. NOTES

6.1 Intended use. The A/S32A-16 crane is intended for use in handling completely assembled missiles within its capacity or components as required during assembly and pre-launching operations.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Location and conditions for preproduction testing (see 4.3).

c. Level of preservation and packaging, and packing (see 5.1 and 5.2).

6.3 Definitions. For the purpose of this specification, the following definitions apply:

6.3.1 Reliability. Reliability is defined as the probability of performing a specified function under given conditions without failure for a specified period of time. Recognizing that, in general, the rate of failure of equipment is fairly constant throughout the life of the equipment, the probability of nonfailure over an operating time interval decreases exponentially as a function of the length of the interval, during which time there is a constant failure rate, and can be expressed as follows:

Confidence level: 90 percent.

$$\frac{\text{Total Test Time}}{2.3}$$

$$\text{Reliability} = e^{-x}$$

$$\text{where } x = \frac{\text{Mission time}}{\text{MTBF}}$$

The $2.3 = \frac{4.61}{2}$ is based on the constant for the Poisson/Chi squared distribution, assuming an exponential (2 degrees of freedom) distribution even though the failure rate for the test time is zero.

6.3.2 Mean-time-between-failures. The mean life or mean-time-between-failures is the arithmetical mean (average) of the operating time between failures.

6.3.3 Normal working cycle. A normal working cycle shall be through 270° rotation. The crane is capable of a continuous rotation of 360°. However, due to interference with the carrier structure, maximum lifts should not be attempted over the front, 45° to either side of the longitudinal centerline.

6.3.4 Safe load. Safe load shall be a load that is not more than 85 percent of the tipping load when tipping is a controlling factor, for a given radius in the least stable position with the vehicle resting on firm, level ground, fully serviced, and with the fuel tanks not less than 1/2 full.

6.3.5 Tipping. Tipping shall be that condition of stability of the vehicle when, at a particular radius and load, all tires of a single wheel (when the vehicle is used without outriggers) opposite the load leaves the ground by at least 1/4 inch, but not more than 4 inches or when outriggers are used, at a particular radius and load, both outriggers opposite the load leave the ground by at least 1/4 inch.

6.3.6 Tipping load. Tipping load shall be the weight required to cause tipping at any specified radius where tipping is the controlling factor. It shall consist of the hook block slings and other load weight components attached to the hook of the hook block and shall not include weight of crane boom, wire rope, hook block, or other components necessary for crane operations.

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6.3.7 Boom radius. The boom radius shall be the horizontal measured distance from a point on the centerline of rotation of the revolving superstructure to the centerline of the crane hook block with load suspended.

6.3.8 Rated safe load. The rated safe loads shall include all auxiliary load handling devices, such as load falls, slings, load block and hooks.

6.4 Preproduction sample. The sample may be a preproduction item or a production item which meets the requirements of this specification. In either case, the sample shall be identical with the proposed item and shall be made or shall have been made by the methods that are to be used or were to be used in production of items to fulfill the terms of the contract. The preproduction sample may be shipped as part of the contract. Approval of a preproduction sample shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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